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3 **Energy and Water Dynamics in Data Center Cooling: Insights from a Modeling Study in Hot-Arid Climates**
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Highlights:

► Modeled hourly energy and water use of cooling systems in Phoenix, AZ data centers

► Compared energy and water use and savings of economizer-based cooling systems

► Economizers and evaporative cooling save energy and water in limited hours

► Adding pre-cooling to air-cooled chillers saves energy but increases onsite water use

Abstract

Data centers (DCs) growth has resulted in increased energy and water usage for computer room cooling, influenced by climate conditions. This study developed 8760-energy and water models to estimate the hourly energy and water use in Phoenix, AZ DCs, and examined three cooling systems (air-cooled chillers, water-cooled chillers, and evaporative cooling) with a particular emphasis on seasonal effects. The models also estimate the energy and water savings using air-side and water-side economizers (ASE and WSE) year-round. The air-cooled chillers consume the most energy and source water annually, while water-cooled chillers have the largest annual onsite water use. Utilizing ASE results in approximately 22% energy and source water savings for air-cooled chillers and 29% onsite water savings for water-cooled chillers, contributing to reduced environmental impacts. Integrating pre-cooling effects in air-cooled chillers results in 11% annual energy savings. Chillers can be operational throughout the year, while evaporative systems integrated with ASE provide cooling for only 64% of the year. With computer room cold aisle containment (CAC), ASE can cool the DCs for approximately 35% of the year, while WSE can cool the DCs for approximately 12% of the year. These findings provide valuable insights for optimizing energy and water usage in DC cooling that inform future DC design and operation, with a focus on minimizing environmental impacts and promoting sustainability.

Nomenclature

Abbreviations

ACC	Air-cooled chillers
AEC	Adiabatic evaporative cooling
ASE	Air-side economizer
ASHRAE	American Society of Heating, Refrigerating, and Air-Conditioning Engineers
CRAH	Computer room air handler
DC	Data center
EWIF	Energy-water intensity factor
PCACC	Pre-cooled air-cooled chillers
PUE	Power usage effectiveness
RH	Relative humidity
TMY	Typical meteorological year
WCC	Water-cooled chillers
WSE	Water-side economizer
WUE	Water usage effectiveness

Parameters

H_t	Humidity ratio
$\dot{m}$	Air flowrate
P	Power
Q	Heat load

T	Temperature
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Symbols

η	Efficiency
ΔP	Air pressure difference across pump/fan

Subscripts

OA	Outdoor air
OA,Evap	Outdoor air after passing evaporative cooler
SA	Supply air
w	Water

1. Introduction

The increasing rate of data processing and storage demands and advances in the computing and electronic industry have resulted in rapid growth in the global cloud computing industry. Koomey et al. (2020) reported that the energy consumption in global data centers (DCs) increased from 194 TWh in 2010 to 205 TWh in 2018, which accounts for nearly one percent of the global electricity consumption [1, 2]. The global electricity consumption is

also predicted to increase by 15 times, reaching nearly 3,000 TWh by 2030; this estimate corresponds to between 3% (best-case) and 13% (worst case) of the projected global electricity demands [3]. Additionally, during the global COVID-19 pandemic, some countries reported more than a 20% increase in internet demands to accommodate the recently developed “digital lifestyle,” highlighting the increasing trend of DC energy use around the world [4].

Given the exponential growth of DCs and their increasing environmental impacts, it becomes important to address the water and energy needs of DCs [5, 6]. The water-energy nexus plays a crucial role because of the interdependence between water and energy in cooling processes within DCs [7]. Cooling systems, particularly computer room conditioning, are instrumental in maintaining the optimal temperature for IT equipment and ensuring their reliable operation. However, the computer room air conditioning process can be energy intensive, accounting for 30-40% of the total DC input energy [8-11]. Understanding and addressing the water and energy requirements and their nexus are crucial to improving the energy and water usage efficiencies and mitigating environmental impacts.

Energy consumption from server and cooling operations accounts for the largest shares of energy use in DCs [12-14]. Energy consuming components in DCs include IT equipment, computer room air cooling systems, uninterruptible power supplies (UPS), power distribution units (PDU), switches, and light sourcing. Specifically, IT equipment comprises nearly 50% of DC electricity use, where almost all the energy consumed is converted to low-grade, or waste heat. Therefore, cooling systems are critical to remove the generated waste heat to prevent potential IT equipment failure.

1.1. DC cooling systems

Cooling systems in DCs can be widely classified into three main categories: 1) refrigeration cycles, 2) evaporative cooling, and 3) free cooling or economizers (Figure 1). In recent years, liquid-cooling has gained attention in high-performance computing centers with high rack densities. Liquid-cooling usually necessitates additional infrastructure and specific IT equipment because of the direct exposure to cooling fluid and is therefore mainly limited to use in enterprise DCs [15-17].

76 Refrigeration cycles are common cooling technologies used in DCs and are energy intensive. Two of the most
77 common refrigeration cycles include direct expansion (DX) and chillers/vapor-compression. DX systems utilize
78 computer room air conditioner (CRAC) units to cool the computer room air and air-cooled condensers to reject the
79 waste heat to the outdoor environment. In chilled-water cooling plants, computer room air handler (CRAH) units
80 are used to cool the computer room air and reject the waste heat. The waste heat is transferred to the returned
81 chilled water and rejected to the outdoor environment by either air-cooled condensers (air-cooled chillers) or
82 water-cooled condensers and cooling towers (water-cooled chillers). Air-cooled chillers use outdoor air for heat
83 rejection and are energy-intensive (nearly zero onsite water use); whereas, water-cooled chillers use cooling tower
84 water for heat rejection and are water-intensive.

85 Evaporative cooling is another low-cost cooling technology used for DC cooling [18]; however, they can be water
86 intensive and not applicable in hot-humid climates because of the high outdoor air wet-bulb temperature [19].
87 Evaporative cooling systems use the latent heat of vaporization to cool computer room supply air. Computer room
88 supply air is cooled by directly misting or indirectly by passing through a water-saturated medium. Evaporative
89 cooling can be direct or indirect [20, 21]. The indirect configuration surpasses direct evaporative coolers in
90 performance by omitting the addition of humidity and outdoor pollution to the supply air.

91 Free cooling systems, also known as economizers, are generally not stand-alone cooling systems and require
92 coupling with mechanical or evaporative systems. There are two types of economizers: air-side economizers (ASE)
93 and water-side economizers (WSE) [22-24]. In ASE mode, after passing through high-efficiency particle filters, the
94 outdoor air is either directly used as computer room supply air, or indirectly used to cool computer room supply air
95 via heat exchangers [25]. ASE systems are most efficient when outdoor air temperatures are lower than the
96 American Society of Heating, Refrigerating, and Air-Conditioning Engineers (ASHRAE) recommended computer
97 room temperature, ranging from 18 °C to 27 °C for dry-bulb and -9 °C to 15 °C for wet-bulb temperatures with RH
98 about 60% (Fig. A-1). In 2015, ASHRAE expanded the environmental envelope by increasing the minimum dew
99 point temperature from 5.5 °C to -9 °C, resulting in expansion of the potential for use of free cooling in DCs [26]. In
100 WSE mode, cold water is used to indirectly cool the chilled water, which is then used for cooling computer room

air. The WSE mode is efficient when a source of cold water, such as cooling tower or natural cold water resources is available [27, 28].

1.2. DC cooling in hot and arid climate

Previously published work on cooling system energy use in DCs located in hot and arid climates (ASHRAE climate zone: 2B) suggests that the use of direct evaporative cooling systems are best-suited for minimizing cooling systems energy consumption because of the efficient evaporative effect induced by the large difference between the dry- and wet-bulb temperatures (e.g. average hourly difference of 9.7 °C) [29-31]. However, some studies suggest that compressor-based cooling systems integrated with ASE are better suited for DCs located in water-stressed areas because of the minimal water and energy consumption with the use of “free cooling” [11, 32, 33]. For example, Lee et al. (2013) demonstrated that for DCs in 2B climate zones, utilizing ASE and increasing the ratio of outside air to the computer room supply from 0 to 1, resulted in about a 14% decrease in energy consumption [32]. Furthermore, the study concluded that ASE can be utilized for over 5,000 hours per year, resulting in nearly 21% energy savings compared to stand-alone water-cooled chillers [32]. Another study by Gozcu et al. (2017) found that direct ASE cooling requires the least amount of energy among various cooling methods, followed by indirect evaporative cooling, indirect ASE, and WSE [34]. Moreover, a study conducted by Lawrence Berkeley National Laboratory indicates that ASE energetics outperforms WSE in DCs [35]. Despite the lower energy savings of WSE compared to ASE and indirect evaporative cooling, WSE offers additional advantages, such as the practicality and ease of retrofitting with minimal disturbance to airflow, especially in larger DCs [34, 36].

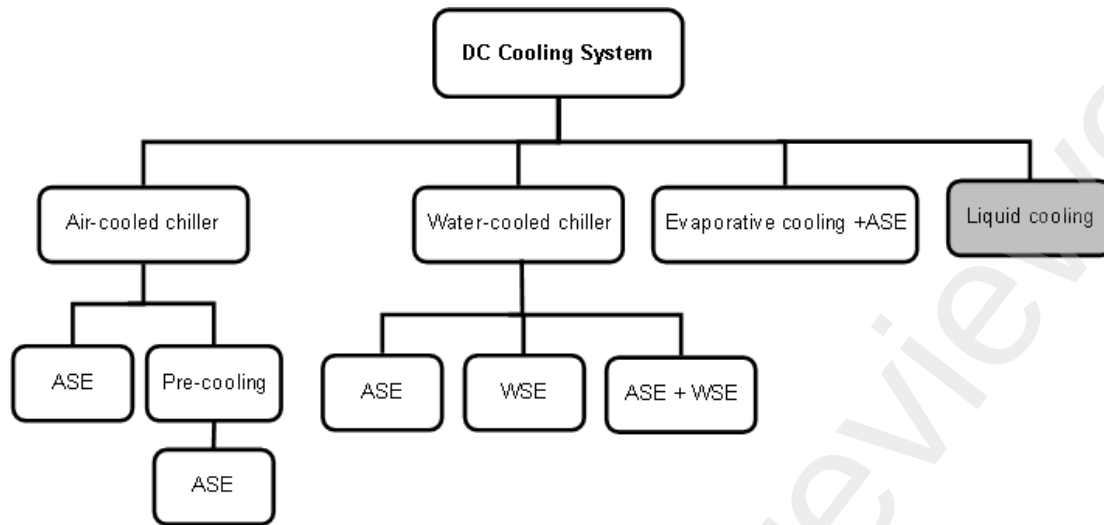


Figure 1. Categorized commonly used cooling systems in DCs, including mechanical and evaporative cooling (with and without integration with economizers), and liquid cooling. Liquid cooling requires unique infrastructure and server requirements and is out of scope of this study. ASE and WSE stand for air-side economizer and waterside economizer, respectively.

Chiddix and Zion found that in the Phoenix climate type (2B: hot and dry), air-cooled chillers with 3.3 MWh/kW IT load consume three times more annual energy than water-cooled chillers and 22 times more annual energy than evaporative cooling systems. Additionally air-cooled chillers with use 2.6 times more total water (91,310 m³/MW IT load onsite and source water for power generation) than water-cooled chillers [37]. The study assumed a constant chiller efficiency despite the large effect of outdoor air temperature on chiller efficiency and did not evaluate the use of economizers [38-40]. The total water consumptions were also estimated using an energy-water intensity factor (EWIF) of 27.4 L/kWh, higher than previously reported EWIF values for the area ranging from 1.21 to 1.62 L/kWh [7, 41].

Cooling-related variations of normalized power and water use in DCs are influenced by regional climate, cooling configurations, the size of the DC, and the corresponding operating practices adopted [42, 43]. Lei and Masanet examined the variation in annual power usage effectiveness (PUE) and water usage effectiveness (WUE) across three different categories of DC size classifications (i.e., large, midsize, and small) based on taxonomy proposed by Shehabi et al. [44]. The study found that in almost all climate zones, including zone 2B, the average value and

standard deviation of PUE and WUE in DCs increased with decreasing DC size; this observation can be attributed to lower air flow efficiencies and different operating practices that are commonly adopted in smaller DCs [42].

While numerous published studies provided valuable insight into the annual energy and water use of DCs in different climate types, there is a notable lack of investigation into the effects of seasonal changes on energy and water use (both onsite and source) in DCs. Addressing this gap is essential for DC designers to comprehend the patterns of energy and water use throughout the year, leading to a better understanding of potential opportunities for energy and water savings. By comparing and identifying the optimal cooling solutions, designing hybrid cooling systems (e.g. integrating pre-cooling effects in air-cooled chillers), and utilizing economizers year-round, large energy and water conservation can be achieved.

Furthermore, despite the development of standards and metrics by the DC industry to quantify energy and water consumption, there is considerable variation in cooling systems used among DCs in different climate types. Unfortunately, little consideration has been given to local resource constraints. In hot, arid climates like the U.S. southwest, the water demands of various cooling systems differ from those used in cooler climates. Cooling towers, in particular, in DCs similar to coal-fired power plants, require large volumes of water for makeup water. Consequently, DCs in the southwest region are facing increasing pressure from local communities to conserve limited freshwater supplies by adopting alternative water sources and reducing freshwater consumption and withdrawals. Cooling systems have unique operating parameters that maintain desired IT conditions. Often, these parameters are influenced by seasonal variations, climate type, and cooling system type. Understanding these relationships, particularly in water-stressed regions such as Phoenix, AZ, can contribute to significant energy and water conservation. Despite prolonged drought conditions, the greater Phoenix area has become the fifth-largest DC market in the US, with a capacity of 280 MW in 2022, indicating a growth rate of about 11 times compared to 2021 [45, 46]. Therefore, it is critical to investigate the water and energy nexus, identify potential areas for water and energy savings, and consider local resource constraints on DC cooling systems in resource stressed areas, including Phoenix, which is critical for continued growth of more resilient and environmentally responsible DCs in the future.

The objective of this study is to develop 8760-energy-water predictive models for three distinct cooling systems with and without economizers ($n=9$) for DCs located in hot-arid climates such as Phoenix, AZ. We hypothesize that a direct trade-off exists between water and energy consumption in DC cooling systems. By implementing hybrid-cooling systems that take advantage of environmental conditions and local resource constraints, we aim to improve the overall water and energy efficiency in DCs and promote their sustainable operation of DCs with minimal adverse environmental impacts. By simulating hourly DC energy and water usage over a year (8760 hours) for the nine different cooling system permutations, energy and water use can be compared and potential savings be identified. Modeling results can inform stakeholders and utilities about energy and water savings using hybrid cooling systems, based on daily and seasonal variations.

2. Methodology

An 8760 analysis was conducted to model the energy and water usage by different cooling technologies in DCs located in Phoenix, AZ (2B climate zone). The hour-by-hour analysis method was employed to develop energy and water models of cooling scenarios for all 8,760 hours over a 12-month period. By incorporating sequencing weather data and other design and operating factors, this method provides an applicable approach for improving the design and operating accuracy. First, the energy and water requirements for an IT load of 5.3 MW IT was calculated using the developed models (base on the IT capacity of a data hall in a current Phoenix DC [7]). Then, to facilitate comparison, the results were normalized based on one MW IT load. It was assumed that the uninterrupted power supply (UPS) and power distribution units (PDU) were placed outside the computer room; therefore, the calculated heat load excludes the heat from power loss in these units.

2.1. Meteorological data acquisition

Typical meteorological year (TMY3) weather data was obtained for Phoenix-Sky Harbor International Airport from HandsDown software [47]. The TMY3 dataset provides psychometric data for a one-year period, representing the average 30 years of weather data.

2.2. Cooling systems configurations

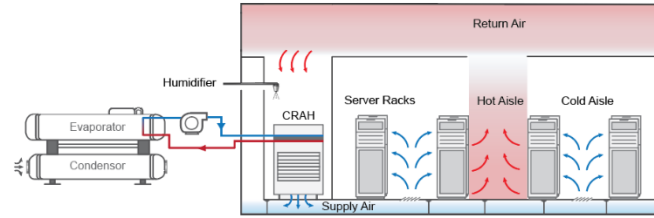
Three distinct cooling systems, including air-cooled chillers, water-cooled chillers, and adiabatic evaporative cooling systems with and without economization ($n=9$) were modeled (Fig. 2). Cooling systems include: stand-alone chilled water systems (Figs. 2a, 2c, and 2e, respectively); ASE integrated with air-cooled chillers, pre-cooled air-cooled chillers, and water-cooled chillers (Figs. 2b, 2d, and 2f); water-cooled chillers integrated with WSE (Fig. 2g); water-cooled chillers integrated with both WSE and ASE (Fig. 2h); and adiabatic evaporative cooling integrated with ASE (Fig. 2i).

2.2.1. Economizer operation

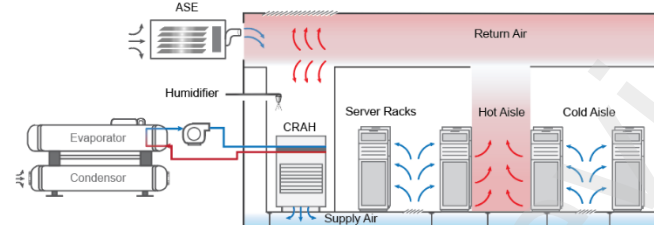
When economization mode is enabled, the priority for handling cooling is given to economization mode over the primary cooling system. Therefore, when the economizer mode is activated, the cooling load on the primary cooling system is reduced to zero. To assess the feasibility of implementing an ASE mode with a chilled water plant and evaporative cooling in Phoenix climate conditions, two scenarios are considered: (1) using outdoor air exclusively; (2) partially mixing outdoor air and computer room return air. Under the first scenario, outdoor air is utilized for cooling only under favorable outdoor air conditions: outdoor air temperature (T_{OA}) is 18.9°C and relative humidity (RH_{OA}) is within 50% to 80% [42, 48, 49]. In such circumstances, the outdoor air is utilized to its fullest potential for computer room cooling, while circumventing the need for additional cooling and humidification. However, when T_{OA} is lower than the computer room supply air temperature setpoint (T_{SA}), and RH_{OA} is lower than the recommended range of supply air relative humidity (RH_{SA}), outdoor air is partially mixed with the computer room return air (second scenario).

While all cooling configurations can incorporate ASE mode, WSE is exclusively used in water-cooled chiller systems that rely on cooling towers to supply cold water. WSE mode availability depends on the temperature difference between the supply chilled water and the cooling tower water [50]. As shown in Fig. 2g, when WSE is available, the cooling tower water is directed to the WSE heat exchangers instead of the chiller condensers.

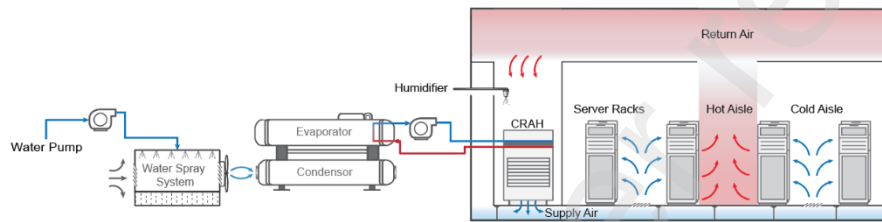
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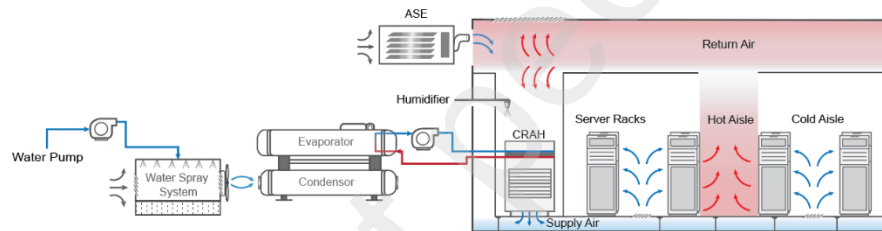
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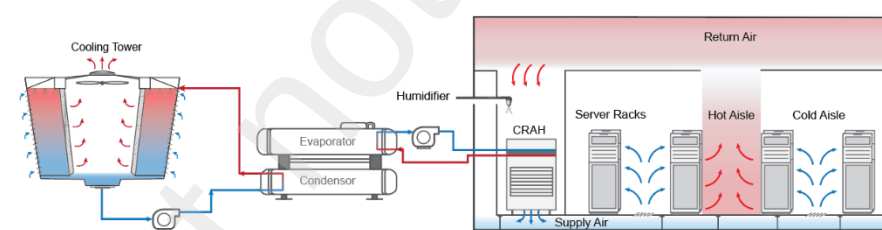
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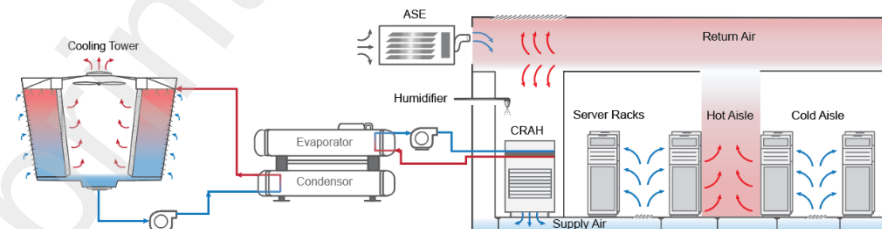
(d)



(e)



(f)



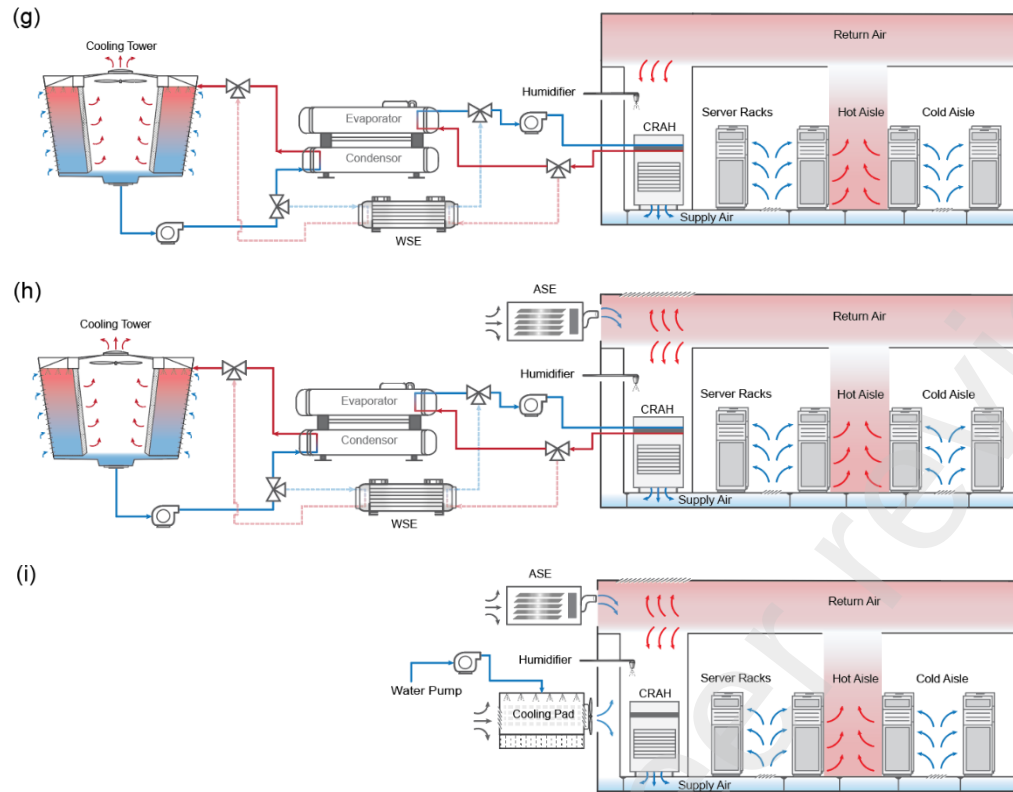


Figure 2. Modeled DC cooling system schematics for (a) air-cooled chillers, (b) air-cooled chillers+ASE, (c) pre-cooled air-cooled chillers (d) pre-cooled air-cooled chillers+ASE, (e) water-cooled chillers, (f) water-cooled chillers+ASE, (g) water-cooled chillers+WSE, (h) water-cooled chillers+ASE+WSE, and (i) evaporative cooling systems+ASE.

2.3. 8760 energy-water analysis model

Excel-based 8760 energy-water predictive models were developed for each cooling configuration based on the 8760-energy analysis provided by Nexant Inc. in previous work on energy auditing service [51]. The original Nexant model focused on evaluating the energy savings achieved by incorporating the pre-cooling effect in the air-cooled chillers, considering only the energy consumption of stand-alone air-cooled chillers and pre-cooled air-cooled chillers (compressor only). However, the model was subsequently enhanced to include additional components, such as pumps, fans, and humidifiers. It also accounted for the energy and water use of different chiller configurations with and without economization. In addition, an 8760 energy-water analysis model was developed

for water-cooled chillers, which includes energy and water use of cooling tower. The energy consumption of chilled water pumps and CRAH units were also incorporated into all models.

The model accounts for the DC cooling load, computer room requirements (i.e., airflow, temperature and humidity conditions), mechanical equipment specifications (i.e., compressor type, size, allowable maximum load, rated power, chilled water setpoints) and hourly meteorological data. Based on these parameters, the model calculates the required number of chillers, CRAH units, chilled water flowrate, computer room return air condition (temperature and relative humidity), and all the energy and water use required for meeting computer room air conditioning setpoints. Moreover, the model can calculate the energy and water use and savings for cooling systems that integrate economizers. Therefore, the overall hourly energy and water use in chilled-water plants is determined by the energy use of the chillers and other required components (e.g., fans, pumps, etc.). The modeling parameters for cooling components are based on literature and manufacturing data and are presented in Table 1.

Table 1. Major inputs in the cooling system models.

Parameter	Value	Unit	Reference
Computer room IT load	4	MW	[7]
Computer room supply air dry-bulb temperature	18.9	°C	[48, 49]
Computer room supply air relative humidity	50-80	%	[42]
Ratio of computer room supply air flow to IT load	0.0614	(m ³ /s)/kW IT load	[49]
Computer room air leakage percentage	13	%	[49]
Chiller compressor type	Screw	-	[52, 53]
Air-cooled chiller size	1,200	kW	[7, 54]
Water-cooled chiller size	1,750	kW	[55]
Supply chilled water temperature	11.7	°C	[49]
Return chilled water temperature	18.3	°C	[56-58]
Chiller changeover load	75	%	[7]

Number of fans per chiller	20	-	[54, 59]
Chiller airflow rate per fan at full capacity	7.5	m ³ /s	[59]
Nominal fan power	2.7	kW	[59]
Minimum outdoor air temperature activating pre-cooling effect	24	°C	[60]
Mist effectiveness of wetted media	85	%	[61]
Mist effectiveness of water mist system (spray)	90	%	[62, 63]
Condenser coil thermal effectiveness	86	%	[64]
Water mist pump pressure for pre-cooling	7000	kPa	[60]
Minimum cooling water temperature entering the condenser	15.6	°C	[65, 66]
Temperature approach in cooling tower	4	°C	[42]
Cooling tower cooling range	7.7	°C	[42, 64, 67]
Cooling tower power fan pressure	250	Pa	[43]
Cooling tower power fan efficiency	77	%	[68]
Cooling tower pump pressure	209	kPa	[42]
Cooling tower pump efficiency	70	%	[69]
Cooling tower cycle of concentration	3	-	[70, 71]
Liquid-gas ratio in cooling tower	0.8	-	[42]
Ratio of drift water loss to cooling tower water flow rate	0.05	%	[72]
Cycle of concentration in evaporative cooling system	2.5	-	[70, 71]
Humidification pump pressure	7000	kPa	[42]
Humidification pump efficiency	70	%	[42]
Chilled-water pump pressure	114	kPa	[43, 69]

Chilled-water pump efficiency	70	%	[69]
CRAH fan pressure	700	Pa	[42]
CRAH power fan efficiency	77	%	[42]
CRAH coil effectiveness	77	%	[73]
Temperature rise across the draw-through fan	1.2	°C	[74]
Duct temperature rise	1.7	°C	[74]
WSE pump pressure	144	kPa	[43]
WSE Pump efficiency	70	%	[42]
WSE heat exchanger approach temperature	2.2	°C	[42, 48]
All pumps and fans motor efficiency	90	%	[75]
All pumps and fans power transfer efficiency	90	%	[75]
Energy-water intensity factor (EWIF)	1.62	L/kWh	[7]

2.4. Energy consumption by cooling systems

2.4.1. Chilled water plant energy use

The electrical energy consumption of the chilled water plant is determined using the flow diagram shown in Figure 3. For the chilled water plants, 1,200 kW screw compression air-cooled and 1,750 kW screw compression water-cooled chillers were selected. These chiller types are commonly used for both air-cooled and water-cooled systems within a nominal capacity range of 245 to 2,600 kW [51, 52]. To determine the number of chillers needed, the DC cooling load and chiller changeover load (i.e. the percentage of nominal capacity at that chillers operate) are taken into account [52, 53]. For detailed step-by-step procedure for calculating chiller compressor energy consumption, please refer to the supplementary information (A-2).

2.4.2. Adiabatic evaporative cooling energy use

In the case of direct evaporative cooling process integrated with ASE (Fig. 2i), when ASE mode is unavailable and T_{OA} is larger than T_{RA} while the humidity ratio ($H_{t,OA}$) is lower than the supply air humidity ratio $H_{t,SA}$, the computer

room return air is rejected to the outdoors, and 100% of the outdoor air is cooled using the direct evaporative cooling to meet the computer room supply air conditions. The flow diagram of the energy use by evaporative cooling system integrated with ASE is shown in Fig. 4.

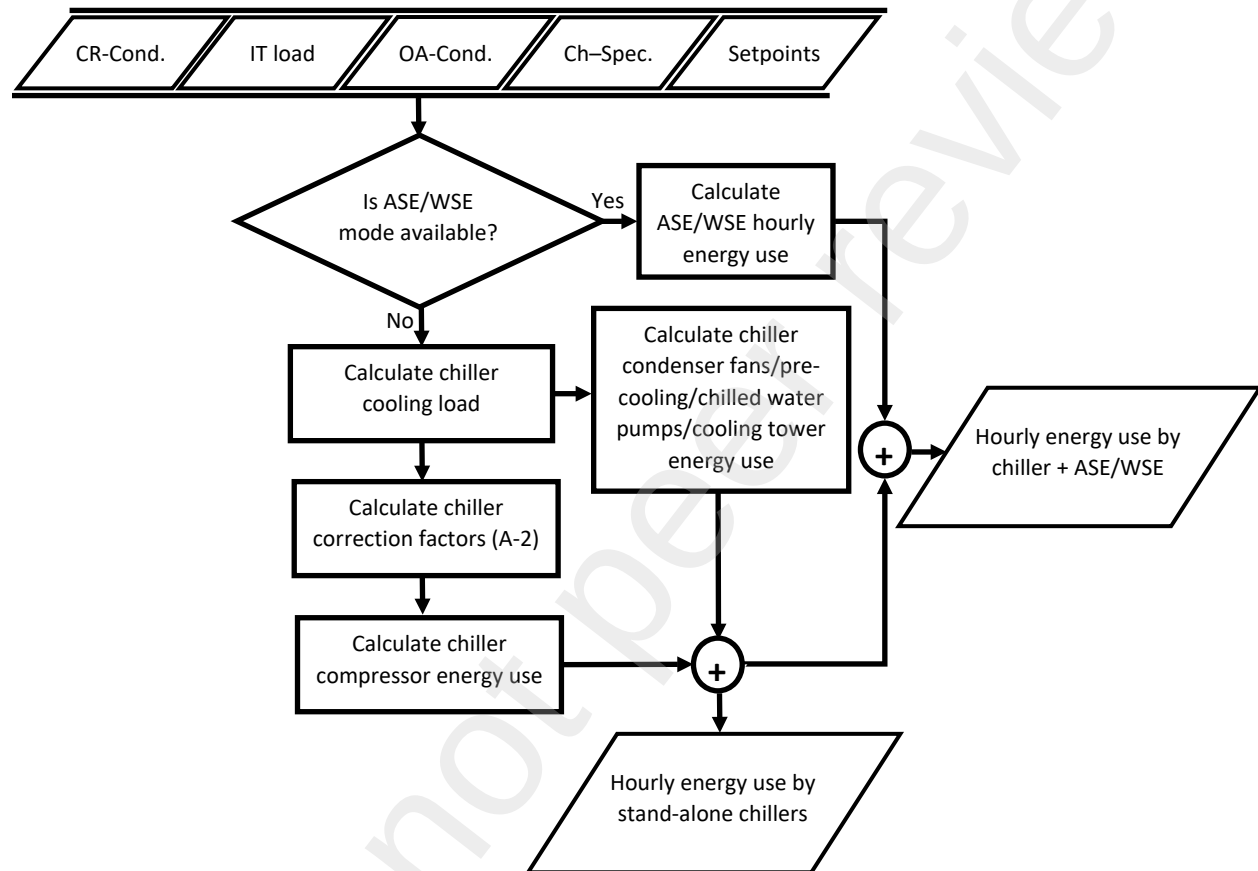


Figure 3. Flow diagram of the energy use by stand-alone chillers and chillers integrated with economizers; where CR-Cond.: computer room condition, including temperature, humidity, and airflow rate; OA-Cond.: outdoor air condition, including air temperature and humidity; Ch-Spec: chiller specification; and Setpoints: operating setpoints for chilled water plant components.

2.4.3. Economization modes energy use

The most power consumptive components in ASE mode are the CRAH fans and the adiabatic humidification pumps. However, when utilizing WSE, the cooling tower pump and fan, chilled water pump, WSE pump, and CRAH fans become the primary power consumers.

The electrical power consumption of fans (P_{fan} , W), is determined by the airflow (Q_{fan} , m³/s), pressure difference across the fan (ΔP_{fan} , Pa), and the fan mechanical (η_{fan}), motor (η_{motor}), and power transfer (η_{trans}) efficiencies. This relationship is represented by Eq. (1) [76].

$$P_{fan} = \frac{Q_{fan} \times \Delta P_{fan}}{\eta_{fan} \times \eta_{motor} \times \eta_{trans}} \quad (1)$$

The electrical power consumption of pumps (PW_{pump} , W), is determined by the pump flow rate (Q_p , m³/s), pump pressure head (ΔP_{pump} , Pa), and pump mechanical (η_{pump}), motor (η_{motor}), and power transfer (η_{trans}) efficiencies, as shown in Eq. (2) [76, 77].

$$PW_{pump} = \frac{Q_p \times \Delta P_p}{\eta_{pump} \times \eta_{motor} \times \eta_{trans}} \quad (2)$$

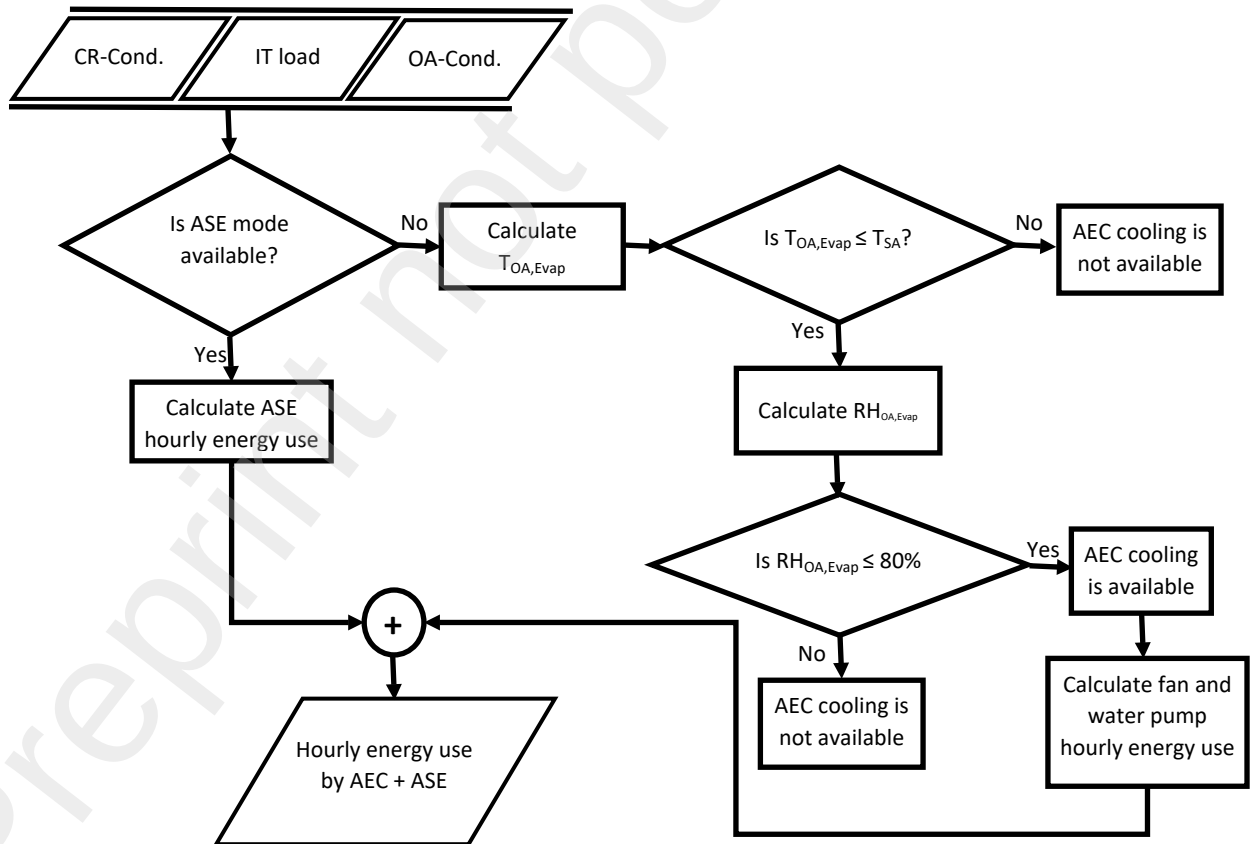


Figure 4. Flow diagram of the energy use by evaporative cooling system integrated with ASE; where AEC: adiabatic evaporative cooling; CR-Cond.: computer room condition, including temperature, humidity, and airflow rate; OA-Cond.: outdoor air condition, including air temperature and humidity; and $T_{OA,Evap}$: effective temperature of outdoor air after passing through the direct evaporative cooling process; and $RH_{OA,Evap}$: relative humidity of outdoor air after passing through the direct evaporative cooling process.

2.5. Onsite water consumption by cooling systems

Cooling systems, including pre-cooled air-cooled chillers, water-cooled chillers, and evaporative cooling systems utilize onsite cooling water for various components, such as pre-cooling compartments (i.e., high pressure water spray pump), cooling towers, and evaporative pads. Furthermore, ASE and WSE modes require water for humidification and cooling tower purposes, respectively.

2.5.1. Chiller onsite water use

The water use by chillers highly depends on the type of chiller. Unlike other types of chillers, the onsite cooling water use by air-cooled chillers is zero because the outdoor air directly enters the condenser. However, the pre-cooled air-cooled chillers utilize water, which is represented by the difference of humidity before and after the pre-cooling effect as shown in Eq. 3. Therefore, the pre-cooling water consumption is determined by the water flow rate ($\dot{m}_w$), dry air flow rate ($\dot{m}_{DA}$), and humidity ratio of air before ($H_{t,in}$) and after ($H_{t,out}$) adiabatic pre-cooling system [42]. The water-cooled chillers primarily consume onsite water for the makeup water of cooling tower, which is necessary because of water loss from blowdown, drift, and evaporation [64, 67, 78, 79].

$$\dot{m}_w = \dot{m}_{DA}(H_{t,out} - H_{t,in}) \quad (3)$$

2.5.2. Adiabatic evaporative cooling onsite water use

Water use in evaporative cooling systems depends on the humidity content of outdoor air before and after the cooling process, air flow rate through the evaporative cooling pads, and the cycle of concentration in evaporative system. The makeup water used in cooling system is a result of evaporation loss and blowdown stream. The evaporation loss in evaporative cooling pads, similar to pre-cooled air-cooled chillers, is calculated using Eq. (3).

However, in the evaporative cooling strategy, $\dot{m}_{DA}$, $H_{t,in}$, and $H_{t,out}$ represent the computer room supply dry air flow rate, and humidity ratio of outdoor air and computer room supply air required humidity ratio, respectively. The blowdown flowrate is also calculated similarly to the blowdown flowrate in cooling towers [42].

2.5.3. Economizer modes onsite water use

In ASE mode, the amount of water use is limited to only what is necessary to provide the required humidity to the computer room supply air. The calculation of water consumption in this process is similar to that of evaporative cooling and pre-cooled air-cooled chillers as shown in Eq. (3). In this equation, $\dot{m}_{DA}$ presents the computer room supply dry air flow rate, $H_{t,in}$ represents the humidity ratio of the mixed air, which consists of a fraction of return air and outdoor air, and $H_{t,out}$ represents the computer room required humidity ratio.

The water usage in WSE mode comprises makeup water used in the cooling tower. However, in this case, the cooling load and circulating cooling water are less than the conditions where water-cooled chillers are used. When water-cooled chillers are in use, the heat generated by the chillers is added to the computer room heat load, while the extra heat load is eliminated when the chillers are not in use, and the cooling load is handled by WSE.

2.6. Cooling systems source water use

In this study onsite water refers to water used onsite at the DC and source water refers to the water used to generate cooling energy required by DCs cooling systems. Equation 4 is used to calculate source water use for all cooling systems using an EWIF of 1.62 L/kWh as indicated by SRP [7].

$$\text{Source water use} = \text{DC cooling power use} \times \text{EWIF} \quad (4)$$

3. Results and Discussion

3.1. Economizers and cooling systems availability

Modeling results of monthly availability for both ASE and WSE cooling modes over a period of one year are shown in Fig. 5a. Overall, ASE and WSE modes can provide cooling for about 3,057 hours and 1077 hours, respectively. The ASE mode is most prevalent during the cooler months (November through April), operating between 8% and 93% of the total hours each month. December marks the peak availability, with ASE mode supplying up to 93% of

325 the cooling required for the computer room. This high availability is because of the lower dry-bulb temperature
326 and humidity ratio of the outdoor air, which enhances the potential for effective mixing with the computer room's
327 return air in ASE mode. However, ASE availability reduces to zero hours during hot months (June through
328 September) because of increased outdoor temperature.

329 While ASE operating results can apply to all cooling systems, the total number of WSE operating hours corresponds
330 to water-cooled chillers integrated with WSE and water-cooled chillers coupled with both ASE and WSE cooling
331 scenarios. The availability of WSE is mostly limited to the cooler months (December through April), ranging from
332 5% to 51% operational hours per month, with the highest availability in January. The observed trend is because the
333 availability of the WSE depends on the temperature of water leaving the cooling tower, which decreases as the
334 outdoor air wet-bulb temperature drops [80]. Therefore, during January, when the monthly average outdoor wet-
335 bulb temperature is at its lowest (6.2 °C), WSE has the highest availability.

336 The availability of evaporative cooling integrated with ASE is shown in Fig. 5b. While cooling systems that utilize
337 refrigeration cycles can meet 100% of the cooling demands throughout the year, the availability of evaporative
338 cooling systems is affected by the outdoor air wet-bulb temperature and is limited to meeting cooling demands for
339 about 64% of the year (5,641 hours per year). The evaporative cooling system limitations are because of the higher
340 monthly average wet-bulb temperature during hot months (June through September) compared to other months
341 [7].

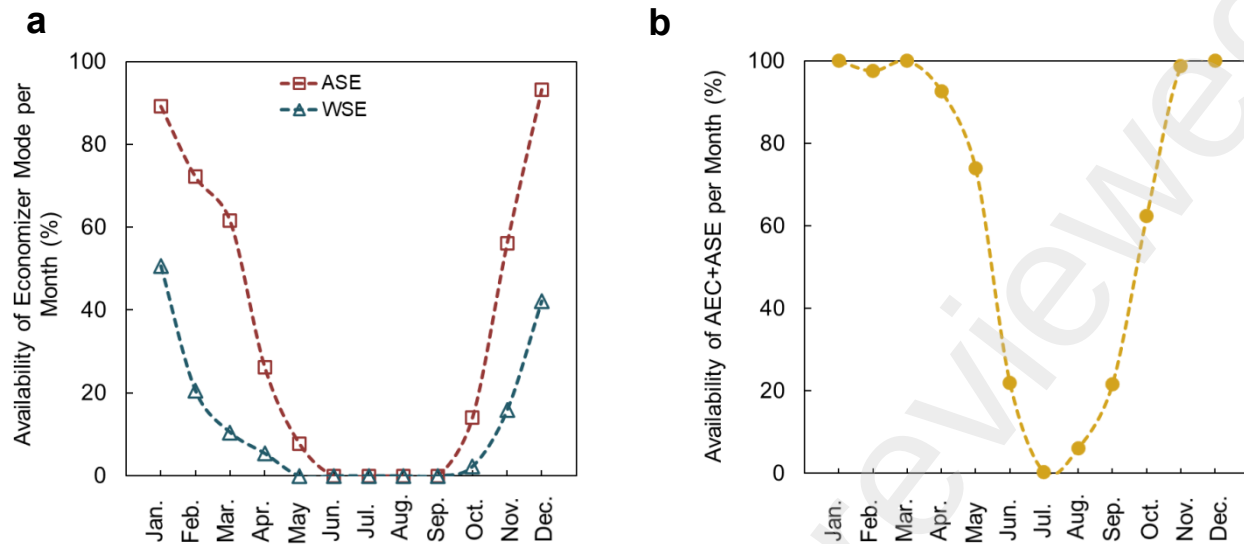


Figure 5. Availability of cooling systems as a function of time for (a) the percentage of hours per month that are cooled by economizers, and (b) the percentage of hours per month that are cooled by evaporative cooling integrated with ASE. AEC stands for adiabatic evaporative cooling.

3.2. Cooling system energy and water consumption

The normalized annual energy and water (onsite and source) usage by cooling systems with and without economizers and annual energy and water (onsite and source) savings for each cooling system integrated with economizers are presented in Fig. 6. The air-cooled chillers have the highest normalized annual energy use of 3.30 GWh/MW IT load. Integrating a pre-cooling effect (evaporative-assisted) in air-cooled chillers reduces the energy consumption of the air-cooled chillers by approximately 11%, as the addition of an evaporative pre-cooling stage decreases the outdoor air temperature entering the condenser. The water-cooled chillers have lower energy use compared to both the air-cooled chillers (23% less) and pre-cooled air-cooled chillers (14% less), as shown in Fig. 6a. The evaporative cooling system integrated with ASE has the lowest energy use, requiring approximately 0.45 GWh to provide cooling for about 64% of the year.

Incorporating economizer modes in all cooling systems results in decreased energy consumption (Fig. 6b). In the case of ASE mode, the normalized annual energy use of the air-cooled, pre-cooled air-cooled, and water-cooled chillers decrease on average by 23%. Water-cooled chillers integrated with WSE mode achieve small energy

savings (~6.3%) because of the limited number of hours WSE can be implemented. Integrating both ASE and WSE modes with water-cooled chillers results in negligible difference in energy savings compared to ASE mode alone; because of longer operational hours of ASE mode, WSE economization has small operational hours. Therefore, when both economizer modes are available, ASE mode is preferred over WSE mode considering the lower energy and water consumption of ASE mode.

The descending trend for the normalized annual source water use (Fig. 6c) are consistent with the observed trends for energy use because the source water use is a function of the energy use and EWIF (constant). As shown in Fig. 6c, the air-cooled chillers exhibit the lowest onsite water use, $<1 \text{ m}^3/\text{MW IT load}$, as the only onsite water consumption is for the re-humidification of the computer room recirculated supply air. The overall humidification water use by air-cooled chillers is minimal compared to other cooling scenarios as it is assumed no water condensation occurs in the computer room. However, when ASE is integrated with air-cooled chillers, the normalized annual onsite water consumption increases to about $1,719 \text{ m}^3/\text{MW IT load}$ because the outdoor air is mixed with return air and re-humidification becomes necessary depending on the level of outdoor air humidity. Conversely, the water-cooled chillers have the highest annual onsite water use of $29,089 \text{ m}^3/\text{MW IT load}$. Although integrating a pre-cooling effect in air-cooled chillers reduces the energy consumption of the air-cooled chillers (Fig. 6a), it results in additional onsite water use of approximately $15,211 \text{ m}^3/\text{MW IT load}$ in chilled water plants (nearly zero for the air-cooled chiller), indicating a tradeoff between energy and water use in cooling systems.

Because of the additional onsite water use required for re-humidification to maintain the desired level of humidity in the computer room supply air, onsite water is sometimes used for air-cooled chillers integrated with ASE. In the case of air-cooled chillers integrated with ASE mode, the additional water requirement results in negative onsite water savings (Fig. 6d). However, with water-cooled chillers integrated with economizer modes, onsite water savings can be achieved. The availability of ASE eliminates water loss in the cooling tower, resulting in onsite water savings up to 29%. While, integrating WSE with the water-cooled chillers results in approximately 2% onsite water savings. In the case of water-cooled chillers integrated with WSE, the additional heat load generated by chillers, which is transferred to the cooling water, is eliminated when WSE is available and the chillers are not in use. As a result, the required cooling water flow rate in the cooling tower decreases by 14%. However, because WSE is

available for a limited number of hours (~ 12% of the year), the reduction in onsite water loss in the cooling tower is about 2%.

In addition to analyzing the annual energy use of the cooling systems, the monthly energy consumption (Figure 7a) and the monthly energy savings resulting from the cooling systems integration with economizers (Figure 7b) are also compared. Cooling systems with a refrigeration cycle generally consume more energy during the hot months (May-September) because of the high outdoor air dry-bulb temperature for air-cooled chillers and high wet-bulb temperature for water-cooled chillers (Figure 7a). The monthly energy use of the pre-cooled air-cooled and water-cooled chillers have relatively small standard deviations, with a standard deviation of approximately 16 and 11 MWh/MW IT load, respectively. In comparison, air-cooled chillers exhibit a standard deviation of approximately 43 MWh/MW IT load. The observed differences in energy use for air-cooled chillers and water-cooled chillers is mainly influenced by outdoor air wet-bulb temperature variations, while the air-cooled chiller energy consumption is primarily affected by outdoor air dry-bulb temperature. Based on hourly meteorological data for Phoenix, the standard deviation of hourly wet-bulb and dry-bulb temperatures are 5.6 and 9.8 °C, respectively.

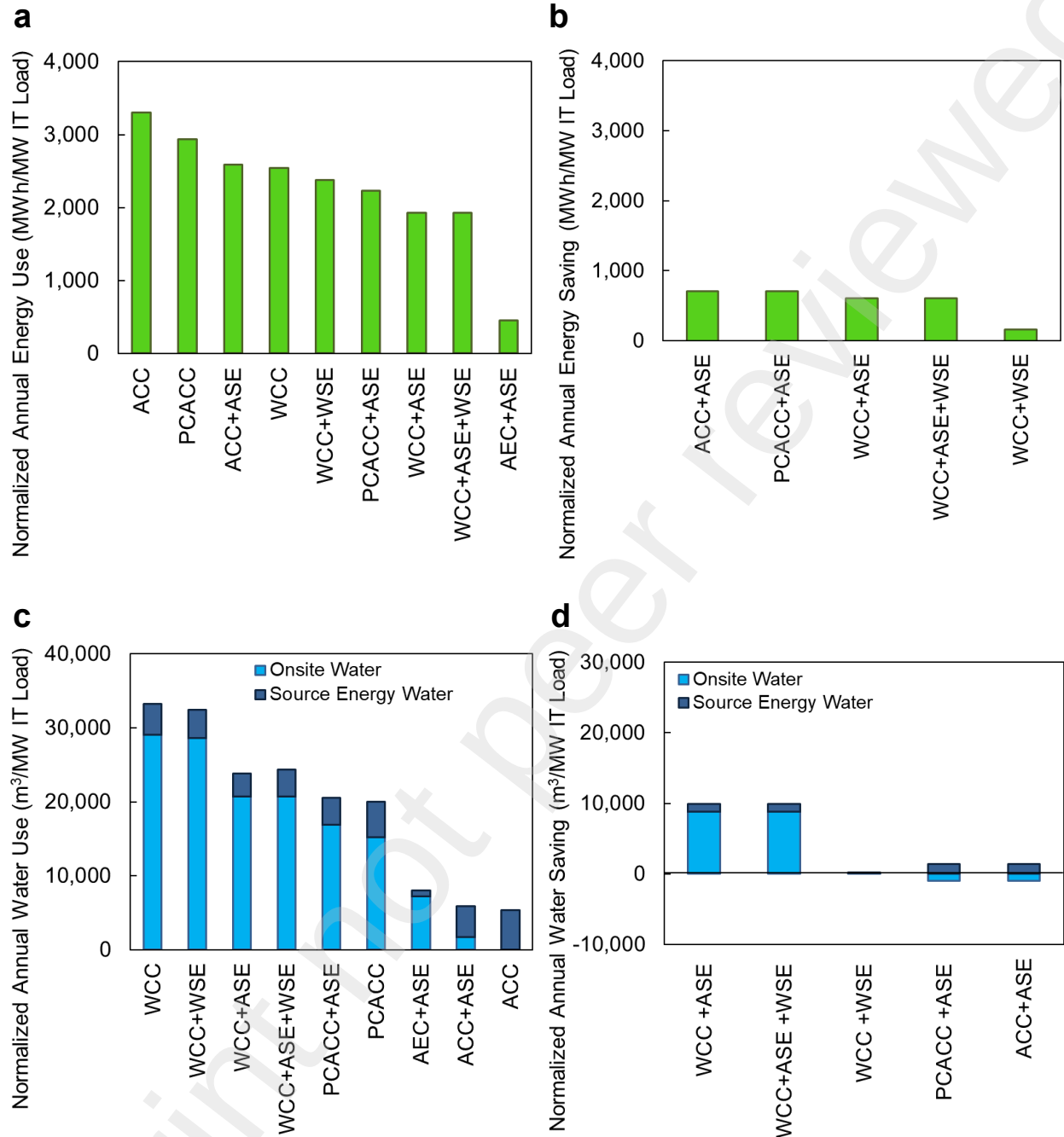


Figure 6. Normalized annual energy and water usage per 1 MW IT load for modeled cooling systems, arranged in descending order of (a) energy consumption, (b) energy savings achieved through economizer modes, (c) onsite and source water usage, and (d) onsite and source water savings achieved through economizer modes. ACC: air-

cooled chiller, PCACC: pre-cooled air-cooled chiller, WCC: water-cooled chiller, and AEC: adiabatic evaporative cooling.

Although the energy use for all mechanical cooling systems has temporal variations, cooling technologies integrated with economizers exhibit larger temporal variations. For instance, the standard deviation in energy use for pre-cooled air-cooled chillers integrated with ASE (76 MWh/MW IT load) and water-cooled chillers integrated with ASE (62 MWh/MW IT load) is smaller than the standard deviation of energy use by air-cooled chillers integrated with ASE (103 MWh/MW IT load). When ASE mode is used, the cooling energy consumption is limited to humidification (ASE only) and CRAH units, which is smaller than energy use by chillers. The monthly energy use by water-cooled chillers and water-cooled chillers integrated with WSE is small because use of WSE is limited (maximum of 51% of month of January). The energy use by evaporative cooling system integrated with ASE remains relatively constant throughout the year because the fan energy (main energy consumer) is mostly constant. However, the energy use by evaporative cooling integrated with ASE drops to nearly zero during the hot months with high wet-bulb temperatures (June through September) because it is an inefficient cooling method under these conditions.

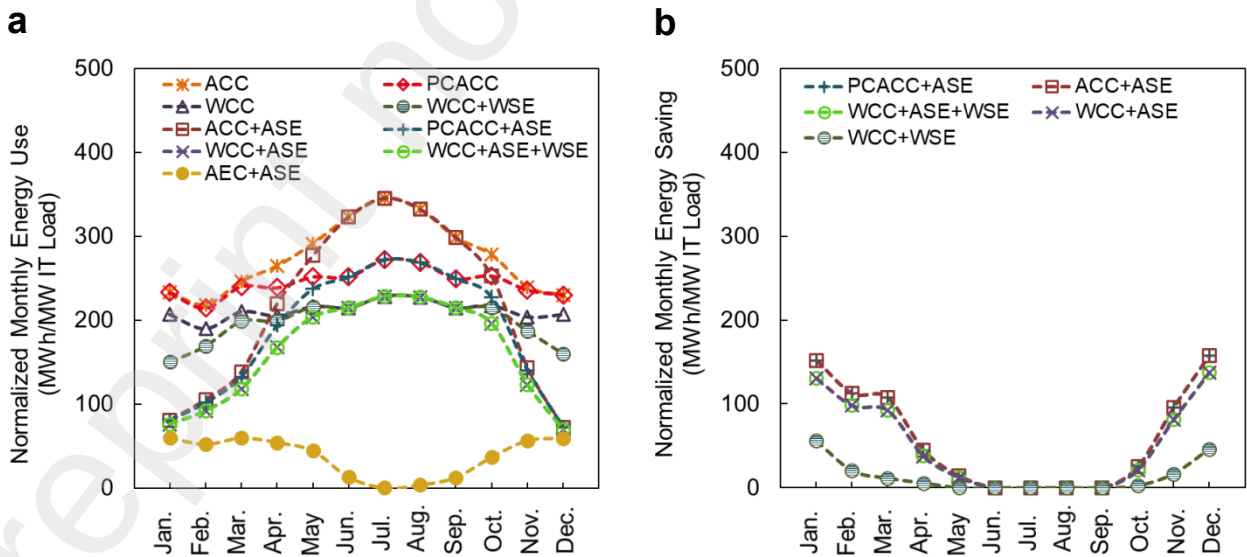


Figure 7. Monthly energy consumption and savings per MW IT load for modeled cooling systems, (a) monthly normalized energy use and (b) monthly energy savings when utilizing economizers. ACC: air-cooled chiller, PCACC: pre-cooled air-cooled chiller, WCC: water-cooled chiller, and AEC: adiabatic evaporative cooling.

The normalized monthly onsite water use and savings for the modeled cooling systems are presented in Fig. 8. Air-cooled chillers have the lowest onsite water use throughout the year because the computer room returned air is used and water requirements for air re-humidification are minimal. The onsite water use of pre-cooled air-cooled chillers increases with rising outdoor temperature, peaking at approximately 2,856 m³/MW IT load in June. Conversely, during the cold months (January and December), pre-cooling water use approaches zero because the outdoor temperature remains below the pre-cooling temperature setpoint (24 °C), rendering pre-cooling unnecessary. In contrast, the monthly onsite water use by water-cooled chillers remains relatively constant through the year, with an average of 2,276 m³/MW IT load and a standard deviation of 92 m³/MW IT load. Water-cooled chillers generally have higher water use than other cooling systems, except in June and July. During these months, the water use by pre-cooled air cooled chillers surpass that of water-cooled chillers because of highest average dry-bulb temperature of 93-96 °C and the lowest relative humidity of 11-22%, which results in higher evaporation rate in pre-cooling process. The observed consistency in monthly onsite water use for water-cooled chillers is attributed to the constant cooling load throughout the year, resulting in similar water loss in the cooling tower every month, with slight variations because of differences in evaporation rates and variation in heat generated by chillers.

During the cooler months (January through April and October through December), when ASE is utilized with air-cooled chillers, the onsite water use increases because of ASE supply air re-humidification requirements. Through the year, the onsite water use by pre-cooled air-cooled chillers with and without ASE are almost similar. However, during the very cold months (January and December) when ASE is the dominant cooling method, covering 89% to 93% of the month, the computer room supply air requires humidification. In general, from January through April and October through December when ASE is available, the pre-cooling effect is not utilized because the outdoor air dry-bulb temperature is mostly below the pre-cooling activation temperature (24 °C). For the rest of the year, when ASE is not available, the pre-cooling effect is active in both pre-cooled air-cooled chillers with and without

ASE, results in similar onsite water use. Water-cooled chillers integrated with WSE exhibit consistent monthly onsite water use, similar to stand-alone water-cooled chillers. When water-cooled chillers operate, the compressors generate additional heat load on the cooling tower, increasing the required cooling water flow rate and water loss compared to when WSE is in use. However, the difference in monthly onsite water use with and without WSE is relatively small, ranging from zero to 7%, with the maximum difference occurring in January when WSE is available for about 50% of the month. Throughout the year, the monthly onsite water use for water-cooled chillers integrated with only ASE or both ASE and WSE is lower than that of stand-alone water-cooled chillers and water-cooled chillers integrated with only WSE, except during the hot months (June through September), when ASE is not available. During the hot months (June through September), when the wet-bulb temperature reaches its highest values (18-21 °C), the availability of evaporative cooling becomes limited, varying from about 0.27% to 22%. Consequently, onsite water consumption drops to its lowest value during this period. However, in the cold months (November through April), the availability of evaporative cooling system integrated with ASE ranges from 93% to 100%, resulting in higher water consumption, with the maximum of 1,346 m³/MW IT load. During this period, the onsite water consumption of evaporative cooling integrated with ASE lies between that of water-cooled chillers integrated with ASE and pre-cooled air-cooled chillers integrated with ASE. This is because during hours that ASE is unavailable and pre-cooling effect is inactive, the evaporative cooling and cooling tower still impose onsite water use, leading to higher onsite water consumption compared to pre-cooled air-cooled chillers with and without ASE.

The onsite water savings associated with economizer modes are presented in Fig. 8b. The observed negative values of onsite water savings for air-cooled and pre-cooled air-cooled chillers integrated with ASE during the cold months (January through April and October through December) is attributed to the water used for re-humidification of the computer room supply air. The largest onsite water savings are observed for water-cooled chillers integrated with ASE during the cold months (December, January, and February), whereas the onsite water savings decreases to zero during the hot months (June through September) when ASE is unavailable.

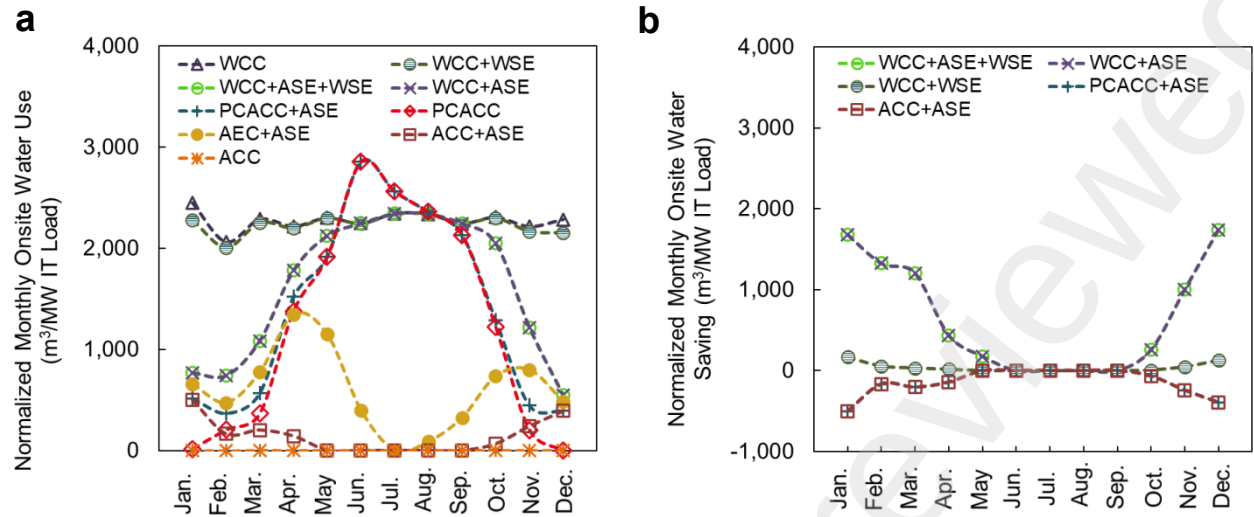


Figure 8. Normalized monthly onsite water use data per one MW IT load of modeled cooling systems, (a) onsite water use and (b) onsite water saving with economizers. ACC: air-cooled chiller, PCACC: pre-cooled air-cooled chiller, WCC: water-cooled chiller, AEC: adiabatic evaporative cooling.

The normalized monthly source water use for different cooling systems and the potential water-saving benefits when cooling systems are hybridized with economizers is illustrated in Fig. 9. The trends for energy use also apply to source water use. Using ASE during the cold season can also reduce source water use. Moreover, utilizing the economizer mode can lead to source water savings for cooling energy (Figure 9b).

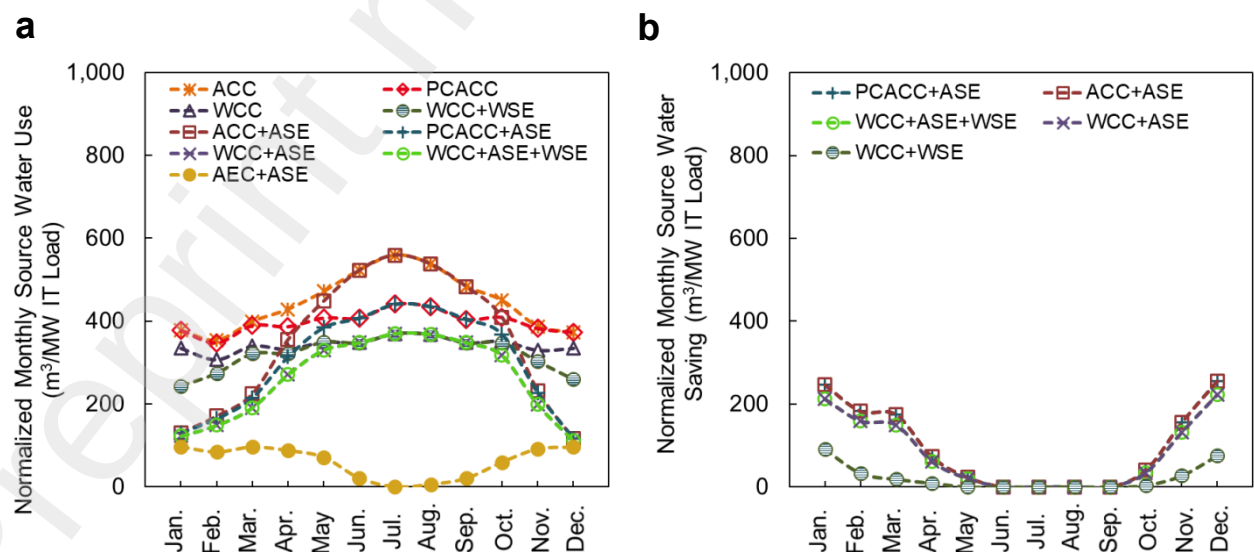


Figure 9. Cooling system monthly normalized source water use and saving per one MW IT load (a) source water use and (b) source water saving with economizers. ACC: air-cooled chiller, PCACC: pre-cooled air-cooled chiller, WCC: water-cooled chiller, AEC: adiabatic evaporative cooling.

Normalized annual energy use by cooling system components is presented in Fig. 10. In all cooling systems, except for the evaporative cooling system, the primary energy consumers are the chiller compressors, while the second highest energy consumers are the CRAH units. However, the energy consumption of the air-cooled chiller compressors decreases when integrated with the evaporative cooling effect (pre-cooled air-cooled chiller). Additionally, the cooling tower fans and pumps in water-cooled chiller plants are other energy-intensive components. Chilled water pumps are the fourth largest energy-consuming components in chilled water plants, particularly in the case of stand-alone chillers. Consequently, the energy consumption of these components decreases when the economizer mode is utilized.

Although the energy use of other components, such as pumps for humidification, pre-cooling/evaporative, and WSE are reported, their value is negligible compared to the other components (Fig. 10). The energy use of evaporative cooling integrated with ASE is smaller compared to the fans and pumps in other cooling systems because of the limited availability of evaporative cooling when integrated with ASE.

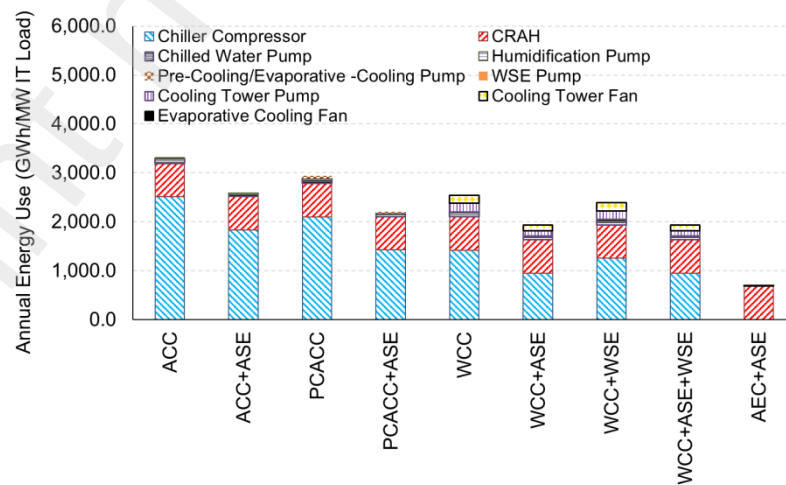


Figure 10. Modeled cooling system component energy use per 1 MW IT load. ACC: air-cooled chiller, PCACC: pre-cooled air-cooled chiller, WCC: water-cooled chiller, AEC: adiabatic evaporative cooling.

4. Conclusion

In this study, energy and water consumption models were developed for various cooling systems suitable for DCs in the greater Phoenix area (2B: hot-arid climate). Models were used to calculate hourly energy and water consumption for an array of stand-alone chillers, including the option of integrating ASE, WSE, or a combination of the two. Based on the results, air-cooled chillers have the highest annual energy and source water consumption, while water-cooled chillers have the highest annual onsite water use. Evaporative cooling integrated with ASE has the lowest energy consumption, but is only effective for approximately 64% of the year. Integrating air-cooled chillers, pre-cooled air-cooled chillers, and water-cooled chillers with ASE resulted in energy and water savings. Furthermore, integrating a pre-cooling effect in air-cooled chillers results in energy savings but increased onsite water use. Therefore, ASE and WSE are generally found to be effective in reducing energy and water consumption in DC cooling systems in the Phoenix climate for a limited number of hours per year. Findings from this research emphasize the substantial potential of adopting hybrid cooling systems and economizers to mitigate the adverse environmental impacts of DCs, making a progress toward enhancing DCs water and energy efficiency and contributing to a more sustainable digital ecosystem.

CRedit authorship contribution statement

Leila Karimi: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Modeling, Funding acquisition, Visualization, Writing – original draft, Writing – review & editing. **Reema Shinh:** Visualization, Writing– review & editing. **Colin Laisure-Pool:** Data curation, Methodology, Writing– review & editing. **Michael Green:** Methodology, Funding acquisition. **Leeann Yacuel:** Funding acquisition. **Jamie Ashby:** Funding acquisition. **Kerri L. Hickenbottom:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Visualization, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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